



Verilog-Based Multi-Car Parking System Using Finite State Machines for Urban Parking Management

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Abstract: The increasing vehicle population in urban areas has intensified parking challenges, leading to traffic congestion and inefficient space utilization. This paper presents a Verilog-based multi-car parking system designed using Finite State Machines (FSMs) to automate parking slot allocation and management. Implemented on an FPGA using Xilinx Vivado 20.1 and simulated in Model Sim, the system supports multiple slots with real-time vehicle tracking via sensors and password-based authentication. By leveraging FSMs, the proposed design achieves low area utilization and scalability, offering an efficient solution for smart parking in urban environments.

Keywords: Verilog, Finite State Machines, Smart Parking, Xilinx Vivado, ModelSim, Urban Parking Management

I. INTRODUCTION

The rapid growth of vehicle ownership in urban centers, particularly in densely populated areas like shopping malls and residential complexes, has exacerbated parking challenges. Limited parking spaces result in traffic congestion, illegal parking, and increased operational costs due to manual management. Traditional parking systems rely on human intervention for slot allocation and access control, leading to inefficiencies and delays. This necessitates automated systems to optimize parking space utilization and enhance user experience. In India, cities such as New Delhi, Hyderabad, and Bangalore, parking shortages are acute, with vehicle ownership outpacing infrastructure development. The economic impact is significant, as parking spaces increase construction costs, affecting urban planning. Poor parking management also contributes to environmental issues through increased fuel consumption and emissions from drivers searching for slots. Smart parking systems, implemented using hardware description languages like Verilog, offer a solution by providing real-time slot availability and automated access control. This paper proposes a multi-car parking system designed in Verilog using an FSM architecture, implemented on an FPGA with Xilinx Vivado 2023.1 and simulated in ModelSim. The system uses sensors for vehicle detection and a password-based authentication mechanism to manage entry and exit, supporting up to four slots with scalability for larger setups. It minimizes manual intervention, reduces area utilization, and enhances efficiency, making it suitable for urban parking applications. The paper is organized as follows: Section II reviews related work, Section III details the system design and FSM architecture, Section IV discusses the existing method's limitations, Section V presents the proposed method, Section VI provides simulation results, Section VII concludes with future scope, and Section VIII lists references.

II. LITERATURE SURVEY

Smart parking systems have gained attention to address urban parking challenges. Traditional systems relied on manual monitoring, which was inefficient due to labor costs and errors. Recent advancements leverage sensors, IoT, and HDLs like VHDL and Verilog for automation. These systems aim to provide real-time slot availability, reduce search times, and enhance efficiency in constrained urban environments. Studies like [1] proposed an FSM-based parking system in VHDL on an FPGA, effective for single slot automation but lacking multi-slot support. Similarly, [2] introduced a centralized system for slot allocation, but its complex infrastructure limits practicality. IoT-based solutions, such as [5], use IR sensors for vehicle detection, offering scalability but requiring reliable network connectivity, which may be challenging in some urban settings. Advanced approaches, such as Fuzzy Logic Control, address these gaps using Verilog and FSMs to create a scalable, low-area multi-car parking solution. By integrating sensor-based detection and password authentication, it balances efficiency and practicality, making it suitable for urban parking facilities with limited resources. The development of smart parking systems has garnered significant attention in recent years, with researchers exploring various technologies to address urban parking challenges. Early systems relied on manual monitoring, which proved inefficient due to high labor costs and human error. Recent advancements have shifted toward automated systems using sensors, IoT, and hardware description languages like VHDL and Verilog. These systems aim to provide real-time parking slot availability, reduce search time, and enhance user convenience, particularly in constrained urban environments. Advanced techniques, such as Fuzzy Logic Control [3] and YOLO-based vehicle detection [7], have been employed to enhance parking system functionality. Fuzzy logic enables dynamic decision making for slot allocation, while YOLO-based systems improve

vehicle detection accuracy but demand significant computational resources. Additionally, [6] proposed an RFID-based system integrated with VHDL, incorporating a recharge mechanism for authentication, yet it faces challenges in cost and maintenance. These works highlight the trade-offs between complexity, cost, and scalability in parking system design.

III. METHODOLOGY

Various approaches have been explored for parking management automation. Reference [1] proposed an FPGA-based system using VHDL FSMs for single slot parking, achieving low latency but lacking scalability. Similarly, [4] used VHDL on a CPLD to prevent indiscriminate parking, but hardware constraints limited its applicability. These systems are effective for small setups but inadequate for multi-slot environments. IoT-based systems have gained traction for real-time monitoring. For instance, [5] employed IR sensors, and [10] used load sensors, both integrated with IoT frameworks. These excel in large-scale settings but rely on stable networks, posing challenges in unreliable environments. Reference [9] proposed cloud-based surveillance, enhancing monitoring but introducing latency and security concerns. The proposed system builds on these works by addressing scalability and resource constraints. Using Verilog and FSMs, it supports multiple slots with a simple sensor-based mechanism and password authentication, achieving low area utilization and cost effectiveness for urban parking applications.

A. System Design and FSM Architecture:

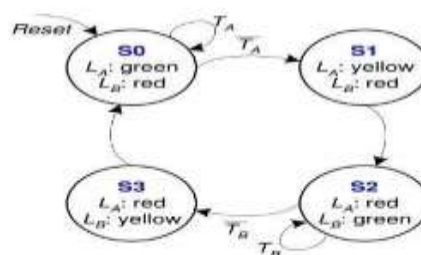
The proposed multi-car parking system employs a Finite State Machine (FSM) architecture to manage vehicle entry, authentication, and exit. Designed in Verilog and implemented on an FPGA using Xilinx Vivado 20.1, the system is simulated in ModelSim for validation. It supports up to four parking slots, with scalability for larger configurations, using a Mealy FSM for efficient state transitions. The architecture ensures robust slot management in urban parking environments. The FSM includes five states: IDLE, WAIT PASSWORD, RIGHT PASS, WRONG PASS, and STOP, as shown in Table I. In IDLE, the system awaits a vehicle, detected by Sensors, transitioning to WAIT PASSWORD for (the password is 10) need to enter correct credentials than RIGHT PASS activating the State Simultaneously.

TABLE I: FSM States and Transitions

State	Description
IDLE	Awaits vehicle detection via 'sensor _{entrance} '.
WAIT PASSWORD	Processes 'password ₁ ' and 'password ₂ '.
RIGHT PASS	Grants access (green light) for correct credentials.
WRONG PASS	Denies access (red light) for incorrect credentials.
STOP	Indicates full parking lot ('car _{count} ' = 4).

FSM State Transition Diagram

- Moore FSM: outputs labeled in each state
- States: Circles
- Transitions: Arcs



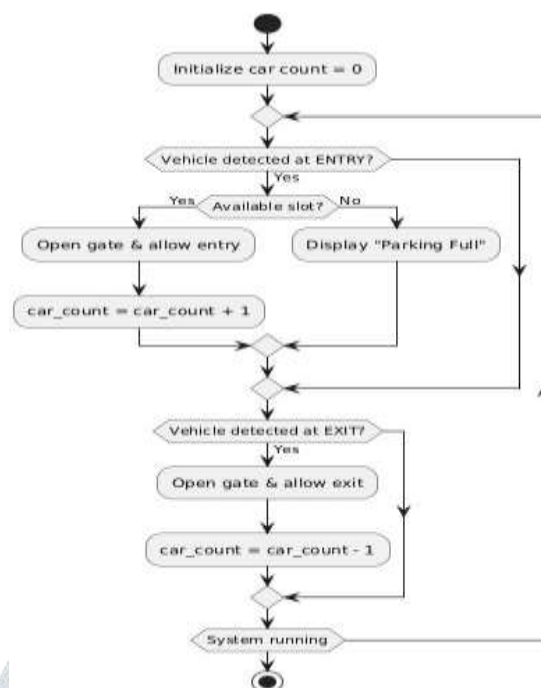


Fig. 1: FSM State Transition Diagram

The FSM flow, depicted in Figure 1, illustrates state transitions driven by sensor inputs and authentication outcomes. The core of the system revolves around a carcount register, which is used to monitor and maintain the count of occupied slots in the parking space. This register plays a critical role in enabling efficient parking management by keeping an up-to-date count of vehicles currently parked in the slot. The system works as follows:

- When a vehicle approaches the entry point, the entry sensor detects its presence.
- The FSM checks the current value in the carcount register to determine if a slot is available.
- If a slot is available (i.e., $\text{carcount} < \text{maximum slot capacity}$), the system:
 - Authenticates the vehicle if required (e.g., RFID, QR code, number plate recognition).
 - Triggers the gate to open for entry.
 - Increments the carcount register by 1 to reflect the newly occupied slot.
 - Updates a display unit or LED indicator to show the updated status (e.g., “1 Slot Occupied”).
- If no slots are available, the system displays a “Parking Full” message and access is automatically denied.”

IV.EXISTING METHOD

Existing parking systems, particularly single-slot designs, are limited in urban environments. Reference[1] proposed an FSM-based system in VHDL on an FPGA, authenticating one vehicle using a sensor and control unit. While effective for small setups, it lacks multi-slot support, leading to inefficiencies in busy parking lots. This restricts its applicability in large facilities like malls or offices.

The block diagram of a typical single-slot system, shown in Figure 2, includes a sensor, control unit, and gate control output. It processes one vehicle at a time, without tracking overall slot availability, forcing drivers to manually search for spaces. This increases wait times and fuel consumption, causing bottlenecks in high-traffic areas.

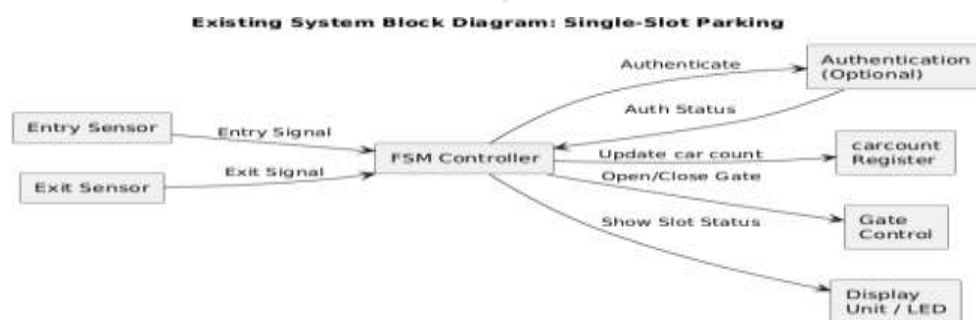


Fig. 2: Block Diagram of Existing Single-Slot System

Limitations include lack of scalability, reliance on manual intervention, and high resource demands. Single-slot systems cannot handle concurrent vehicle movements, requiring human oversight. Many existing methods use resource-intensive technologies like IoT, increasing costs and maintenance challenges. The absence of real-time slot tracking further reduces efficiency in high-demand scenarios.

The proposed system addresses these issues by introducing a multi-slot architecture with real-time tracking, optimized for FPGA implementation. It eliminates manual intervention and reduces resource usage, offering a scalable solution for urban parking management.

V. PROPOSED METHOD

The proposed multi-car parking system overcomes single-slot limitations scalable Verilog-based by implementing a FSM architecture. The proposed system enhances the traditional single-slot parking approach by introducing a multi-slot symmetric FSM-based intelligent urban parking system that tracks multiple vehicles, handles real-time slot allocation dynamically, and offers scalability to multi-level architecture. The method is centered around a robust Finite State Machine (FSM) controller integrated with smart sensors and optimized logic for effective traffic management and resource utilization.

The core of the proposed system is the enhanced car count register logic, now extended to manage multiple parking slots. Each slot is represented as a separate state or a state group within the FSM, allowing for concurrent monitoring of entry/exit events and dynamic updates to availability.

A. Key Features of the Proposed Method

- **Multi-Slot Support:** Unlike the existing single-slot system, the proposed method supports multiple slots (N-slots). Each slot's availability is tracked using a slot status array, updated in real-time as vehicles enter or exit.
- **Advanced FSM Controller:** The FSM manages multiple concurrent transitions and determines slot assignments dynamically. It handles state transitions based on sensor inputs, slot availability, and vehicle authentication status.
- **Symmetric Entry/Exit Handling:** The entry and exit mechanisms are designed symmetrically with their own IR/PIR/Ultrasonic sensors transitions occur independently for each sensor, reducing wait time and enabling bidirectional flow.
- **Slot Allocation Logic:** Upon detecting a vehicle at the entry, the FSM checks for the nearest available slot using a priority encoder-based logic. Once verified, it opens the entry gate and updates the car count and slot status.
- **Display and Notification System:** A dynamic display module is integrated to show real-time availability (e.g., "3 Slots Free") at the entrance. Optional IoT extension can provide mobile notifications or remote monitoring via apps or web dashboard.
- **Security and Authentication:** Vehicles can be authenticated using RFID, QR codes, or license plate recognition to ensure authorized access. FSM integrates this logic as a verification stage before allowing entry.
- **Scalability:** The system is designed to be scalable for larger facilities including multi-level or zoned parking areas. Slot allocation can be localized to zones, with each having a sub-FSM module reporting to a central controller.

The Verilog implementation flow, depicted in Figure 4, outlines the design process: defining FSM states, coding control logic and sensor interfacing, synthesizing with Xilinx Vivado 20.1, and simulating in ModelSim. The Mealy FSM minimizes states, with outputs updated based on state and inputs. The RTL design uses multiplexers for compact logic, validated through simulations for correct/incorrect passwords and full/empty conditions.

The FSM is implemented in Xilinx Vivado with a clock-driven state register for synchronized transitions. The authentication module compares user inputs against predefined values (username: 01, password: 10), ensuring secure access. Synthesis results confirm low resource usage, with multiplexers reducing gate count. This design provides scalability, automation, and efficiency, surpassing single-slot systems for urban parking applications.

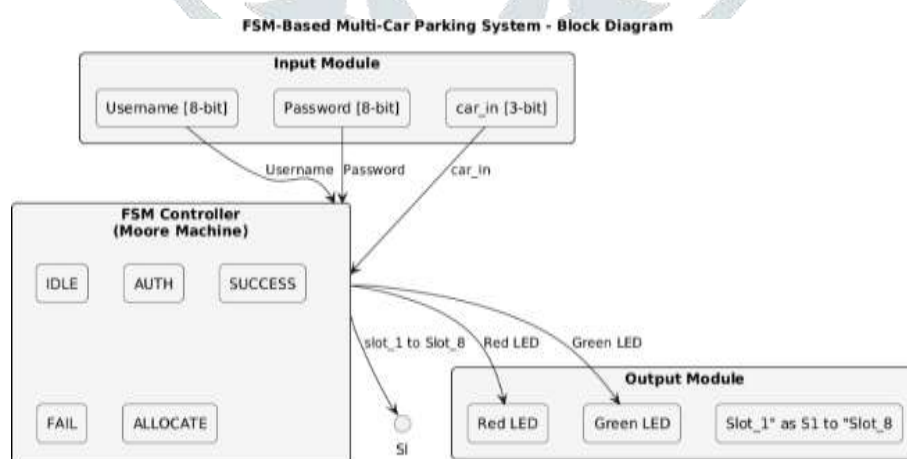


Fig. 3: Block Diagram of Proposed Multi-Car Parking System

The FSM-based multi-car parking system operates through a structured digital control logic that integrates authentication and parking slot management using a Moore finite state machine (FSM). The system receives three main inputs: an 8-bit username, an 8-bit password, and a 3-bit signal `car_in`, which uniquely identifies each incoming vehicle. These inputs are handled by the Input Module, which passes the data directly to the FSM Controller for logical processing. Upon receiving input, the FSM transitions from the initial IDLE state to the AUTH (authentication) state, where it compares the provided username and password against pre-configured valid credentials stored in the design. The FSM architecture is based on Moore principles, meaning that outputs are governed strictly by the current state rather than the instantaneous input. If the input credentials are verified, the FSM enters the SUCCESS state, activating a green LED to indicate successful login. The system then transitions into the ALLOCATE state, where the 3-bit `car_in` input is decoded to activate one of eight available parking slots. Each slot (from Slot_1 to Slot_8) is represented as a separate output signal, and only one slot is asserted high at a time to indicate that the slot is occupied by the current car.

If authentication fails, the FSM transitions to the FAIL state, triggering the red LED to alert the user of the unsuccessful login attempt. No slots are activated in this condition, and the system eventually returns to the IDLE state awaiting the next user input. This strict state-driven behavior ensures that the parking system follows a deterministic and predictable control flow, avoiding conflicts in slot allocation and ensuring secure access management.

The entire logic is implemented in Verilog HDL and verified through simulation in Xilinx Vivado, ensuring the design is synthesizable and testable in an FPGA environment. The absence of sensors or external peripherals allows the system to be used purely in simulation mode, making it particularly suited for educational use, prototyping, and academic research. The modular nature of the FSM also allows easy scalability, enabling the system to be expanded for more users or parking slots in future developments.

VI.SIMULATION RESULTS

The proposed system was implemented in Verilog, synthesized using Xilinx Vivado 20.1, and simulated in ModelSim. Test cases validated vehicle entry/exit, authentication, and slot tracking, ensuring accurate state transitions and minimal latency. The Mealy FSM enables prompt output generation, supporting real-time operation. The system's multi-slot support outperforms single-slot designs, reducing manual intervention.

The RTL schematic, shown in Figure 5, illustrates the synthesized logic with multiplexers and state registers. The design's low gate count conforms efficient resource utilization for FPGAs. The schematic integrates sensor inputs, authentication logic, and outputs, ensuring a compact implementation. Synthesis optimizations in Vivado meet timing requirements for real-time applications.

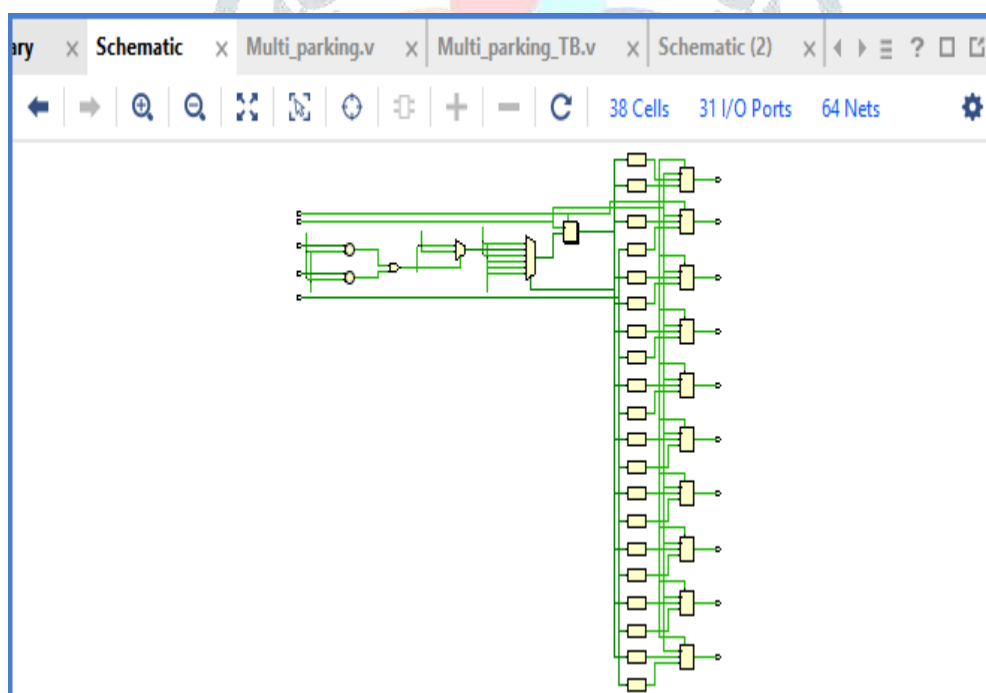


Fig. 4: RTL Schematic of Multi-Car Parking System

The simulation results highlight the system's robustness in handling concurrent vehicle movements and accurate slot tracking. Compared to existing systems, it offers improved scalability and efficiency, validated through comprehensive testing. These outcomes position the system as a practical solution for smart parking in urban environments.

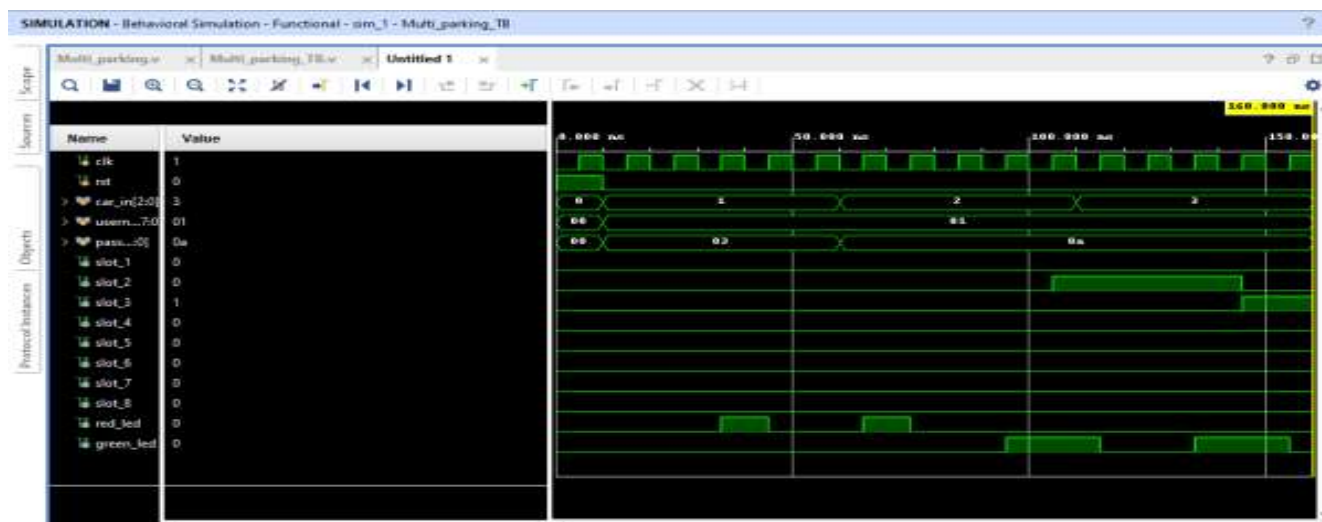


Fig. 5: Simulation Wave form

Implementation results, shown in Figure 6a, detail resource utilization (Look-Up Tables, Flip Flops) and timing analysis, confirming minimal resource usage. The simulation waveform, depicted in Figure 6b, validates state transitions, sensor responses, and outputs (green/red lights) under various conditions, demonstrating reliability and scalability.

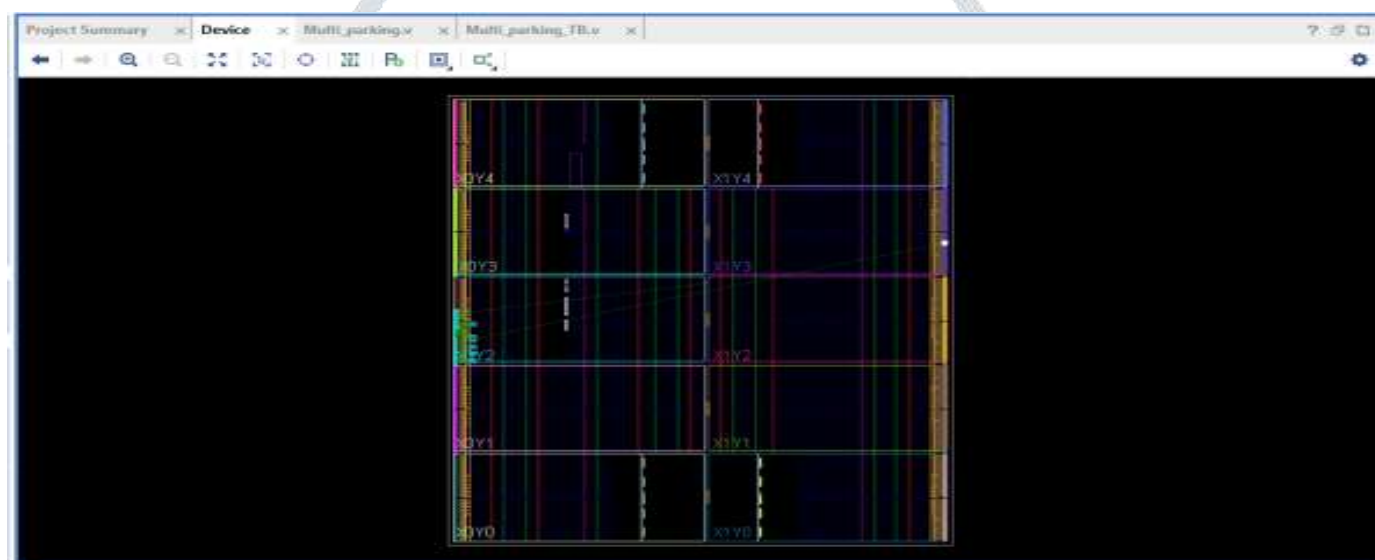


Fig. 6: Implementation

VII. CONCLUSION

The proposed Verilog-based multi-car parking system, utilizing FSMs, effectively addresses urban parking challenges. Its multi-slot support, low area utilization, and automated operation mark a significant improvement over single-slot system. Implemented on Xilinx Vivado 20.1 and simulated in ModelSim, the system ensures reliable performance through accurate state transitions and sensor-based slot tracking. The Mealy FSM optimizes hardware resources, making it suitable for FPGA deployment. Future enhancements could include dynamic authentication mechanisms, such as RFID or database-driven passwords, to improve security. Integrating IoT for real-time slot updates via mobile applications would enhance user convenience. Testing with larger parking configurations and physical sensors (e.g., ultrasonic) could validate real-world performance, ensuring adaptability to diverse urban settings.

Additionally, incorporating machine learning for predictive slot allocation could optimize efficiency in high-traffic scenarios. Exploring hybrid FSM models (Mealy and Moore) enhance performance. Continued research into advanced sensor integration and scalability will position the system as a comprehensive solution for smart parking in smart cities.

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